

Anton Roupasov-Ruiz

Honors Computing Science @ University of Alberta · Software Engineer Intern @ Microsoft

📍 Edmonton, AB, Canada ✉️ roupasso@ualberta.ca 🌐 antonrou.github.io 🐙 antonrou 🌐 antonrou

Education

BSc	University of Alberta , Computing Science (Honors)	Edmonton, AB, Canada Sept 2023 – June 2027
	- cGPA: 3.54 / 4.00 - Relevant coursework: Reinforcement Learning, Machine Learning, Search & Planning in AI, Program Synthesis, Heuristic Search & Artificial Life Research Seminar, Software Engineering Principles, Data Structures & Algorithms, Design & Analysis of Algorithms.	

Industry Experience

Microsoft , Software Engineer Intern	Redmond, WA, USA May 2026 – present
- Engineered a production-grade AI agent integrated with Model Context Protocol (MCP) tools for a new internal platform, letting users translate natural language into dynamically generated, debugged, and deployed enterprise agent workflows. - Architected the backend to orchestrate model invocation via Azure AI Foundry and persist agent conversation histories in Azure Cosmos DB. - Accelerated development cycles by leveraging the GitHub Copilot CLI to manage distributed agent fleets and autonomous subagents.	
Microsoft , Explore (Software Engineering & Product Management) Intern	Redmond, WA, USA May 2025 – Aug 2025
- Identified Bing ETL pipeline bottlenecks with PMs and engineers and implemented the optimal architectural approach. - Built a full-stack tool (React/TypeScript frontend, Java backend) that cut ETL debugging latency from 15 minutes to seconds using Azure Kubernetes Service, Azure Functions, and Azure SignalR Service. - Presented a live demo and quantified impact to stakeholders across Microsoft Bing, including CVPs and VPs.	

Research Experience

University of Alberta — RLAI Lab (Dr. Vadim Bulitko) , Undergraduate Researcher	Edmonton, AB, Canada Sept 2025 – Dec 2025
- Developed the first fully specified, open-source reimplementation of the classic Ackley & Littman (1991) predator-prey evolutionary reinforcement learning A-Life testbed. - Formulated explainable, differentiable decision lists as a programmatic alternative to black-box neural networks, evaluated across 4,000 independent trials with Kaplan-Meier survival analysis. - Proved programmatic agents significantly outlasted neural networks (log-rank $p \sim 4.1e-72$), surviving on average 201.69 steps longer (95% CI [180.14, 223.25]).	
University of Alberta — Computational Psychology Lab (Dr. Jeremy Caplan) , Undergraduate Researcher	Edmonton, AB, Canada May 2024 – Aug 2024
Authored a 4000+ word analysis of four computational associative memory models (grade 4.00/4.00) and implemented MATLAB simulations of neural-network, convolution, local-trace, and matrix models.	

Projects

Prelude — Microsoft Global Intern Hackathon Co-engineered a framework (team of 6) that automatically generates MCP tools for Microsoft Store apps by crawling the OS accessibility tree; demoed at the global hackathon science fair.	July 2026 – July 2026
ERL: Evolutionary RL A-Life Testbed Open-source reimplementation of the Ackley & Littman (1991) A-Life testbed with programmatic, explainable policies that significantly outlived neural networks.	Jan 2025
MentAI Swift-based iOS MVP using the OpenAI API for LLM-assisted wellness insights; pitched to Amii and the UAlberta Health Sciences Startup Incubator.	Dec 2024 – Apr 2025
EEG-Based Pomodoro Focus Timer Python sensor-driven focus timer integrating EEG data streams and signal processing (BrainFlow) for the natHacks 2024 neurotech hackathon.	Jan 2024

Honors & Awards

- Dean’s Honor Roll / First Class Standing — University of Alberta (2025, 2026)
- Jason Lang Scholarship — University of Alberta (2024, 2025)
- Alexander Rutherford Scholarship — Alberta Student Aid (2023)
- AP Scholar with Distinction — College Board (2023)

Leadership

UAlberta Undergraduate AI Society , Panelist, "AI in Industry" Shared industry software-engineering experience at Microsoft.	Edmonton, AB, Canada Sept 2025 – Sept 2025
neurAlbertaTech (UAlberta Chapter) , Treasurer Oversaw finances and the annual budget for a student-led neurotechnology club.	Edmonton, AB, Canada Sept 2024 – May 2025

Skills

Languages: C#, TypeScript, Python, JavaScript, Java, C++, C, MATLAB, HTML/CSS, LaTeX

Tools & Cloud: React, Azure Kubernetes Service, Azure Functions, Azure SignalR, Azure AI Foundry, Cosmos DB, GitHub Copilot, Git, Flask

ML & Data: RL-Glue, Scikit-learn, Pandas, NumPy, Matplotlib

Spoken Languages: English (native), Spanish (native), Russian (native), French (intermediate)